

EXTRAI: six local language models re-extract three published meta-analyses under a perturbation reading proof and a source-verified answer key

Gustavo Barcelos Barra*

September 2026

Abstract

Language models are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from recall of published values, and treat the human answer key as infallible. This pre-registered benchmark ran six quantized open-weight models (8–27 billion parameters) on one consumer mini-computer with an integrated GPU as extractors for three published meta-analyses and their 29 trials: goal-directed fluid therapy (risk ratios), low-carbohydrate diets in type 2 diabetes (mean difference in HbA1c) and methylene blue in shock (odds ratio). The models read only perturbed copies of the trials, with load-bearing numbers displaced under sealed maps so that recitation is detectable, and every cell was graded against a two-layer answer key verified against the primary sources with quotations, which grades the published review on the same ruler. Two extraction sheets ran as a declared arm: 696 sheets, 39.7 h of machine time, no refusals. Cell agreement with the sources ranged from 69% to 90% per sheet on anchor 1, 38% to 71% on anchor 2 and 57% to 100% on anchor 3; the published reviews scored 82%, 100% and 91%. Every sheet rebuilt anchor 1’s pooled morbidity risk ratio within 0.05 of the source-verified value; anchor 2’s pooled mean difference separated the models. Anchor 3’s published estimate (OR 0.73 [0.40; 1.36]) can be reproduced only by reproducing two source-confirmed transcription errors; no sheet reproduced either, and two 14B models scored 32 of 32 cells and pooled to the corrected 0.48 [0.34; 0.69], a mortality benefit where the review reported none. Of 633 non-matching cells, 331 were faithful to the source in another encoding or population layer, 175 were omissions, 106 reading slips and 21 grader artifacts; no attributable recitation occurred. The mandatory-quotation sheet lowered agreement (mean −6.8 cells of 124) and improved no estimate. All materials are public.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, with mature reporting standards [Page et al., 2021], and they are labor-bound: registered reviews take 67 weeks on average from registration to publication [Borah et al., 2017], most of it spent extracting, checking and pooling. Large language models are entering exactly this workflow. Feasibility studies report promising but uneven accuracy for trial data extraction [Schmidt et al., 2024, Khraisha et al., 2024, Konet et al., 2024]; assisted extraction proved noninferior to human-only extraction in a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete reviews in hours [Bakker et al., 2026], with automated drafts rated above a human-authored review by blinded specialists [McLaughlin et al., 2026]. The evidence-synthesis organizations jointly condition such use on demonstrated rigor, human oversight and transparent reporting

*Independent researcher, Brazil. ORCID: 0000-0002-2136-3471. Correspondence: gbbarra@gmail.com.

[Flemyng et al., 2025], and a reporting guideline for studies that use these models now exists [Gallifant et al., 2025].

Three gaps run through that literature. First, most evaluations use commercial cloud models, while clinical text pushes the other way: local open-weight models keep sensitive material on-premises and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. Second, a benchmark that scores agreement with a published review cannot tell a model that *read* the primary trial from one that *remembers* the review: reviews and trials are plausibly in training corpora, memorization scales with model size and duplication [Carlini et al., 2023], and contamination inflates apparent competence [Xu et al., 2024, Golchin and Surdeanu, 2024]. Third, the published review is usually treated as an infallible answer key, although hand-made evidence tables carry their own errors: 17 of 27 meta-analyses using standardized mean differences contained an extraction error in at least one audited trial, some large enough to reverse a finding [Gøtzsche et al., 2007]; extraction and reporting errors were found in most Cochrane reviews of an earlier audit [Jones et al., 2005]; independent observers extracting the same continuous outcomes produced meta-analyses that disagreed [Tendal et al., 2009], in part because trial reports offer several candidate values for one quantity [Tendal et al., 2011]; and the evidence base for extraction standards remains weak [Mathes et al., 2017]. A benchmark that cannot notice when the published value is wrong misgrades its best readers.

However, no evaluation to our knowledge combines the three controls — local models, a reading-versus-recall proof and a key that can overrule the review — on more than one estimand. This study measured six quantized open-weight models on one consumer machine as extractors for three published meta-analyses, under a perturbation reading proof and a source-verified two-layer answer key, and asked whether source-faithful extraction reproduces the published pooled estimates and, where the published cells are wrong, produces the corrected estimate instead.

2 Methods

2.1 Design and registration

The protocol — question, cast, anchors, sheets, hypotheses, instruments, tolerances and compute budget — was registered on 2026-09-08 against a frozen repository commit, reviewed before registration by five independent lenses with a skeptic assigned to each finding (77 findings raised, 43 applied), and not amended afterwards. Corpora, sealed perturbations, answer keys, deterministic engines and graders were built and frozen before the first model call; the models ran on 9–10 September 2026. Six hypotheses were pre-registered (Table 2). The study analyzed only published aggregate reports of randomized trials and three published meta-analyses; no human participants were involved and no ethics approval was required. All materials are public [Barra, 2026].

2.2 Anchors, corpora and answer keys

Three recent meta-analyses served as anchors (Table 1): goal-directed fluid therapy in major surgery [Ashraf et al., 2026], low-carbohydrate diets and HbA1c in type 2 diabetes [Khraise, 2026], and methylene blue in adult shock [Rodríguez-Lima et al., 2026]. Their primary trials — 14, 7 and 8 — formed the three corpora as plain text: article body and tables serialized as text, figures not included (formats in Table 1).

Each corpus was graded against a two-layer answer key. Layer 1 holds the cells the review published; layer 2 holds each cell verified against the trial’s full text with a literal quotation and a per-cell verdict, and it is the grading ruler: where the two layers disagree, the source decides

Table 1: The three anchor meta-analyses and their answer keys.

	Anchor 1	Anchor 2	Anchor 3
Topic	goal-directed fluid therapy in major surgery	low-carbohydrate diet in type 2 diabetes	methylene blue in adult shock
Trials	14	7	8
Corpus format	8 PMC XML; 6 PDF to text	6 XML; 1 PDF to text	4 XML; 3 PDF to text; 1 PDF in Spanish
Estimand	risk ratio: morbidity (4 trials), mortality (2)	mean difference in HbA1c, % (7 trials)	odds ratio, 28–30-day mortality (8 trials)
Published	morbidity 0.778 [0.567; 1.068] mortality 1.021 [0.446; 2.337]	−0.24 [−0.32; −0.16]	0.73 [0.40; 1.36]
Layer 2 (same engine)	morbidity 0.787 [0.580; 1.068] mortality 1.023 [0.446; 2.346]	−0.24 [−0.32; −0.16]	0.484 [0.319; 0.735]
Cells per sheet	124	49	32
Confirmed discrepancies	11 (17-entry record)	0	2 (3 cells; 2 of 8 trials)
Reading proof	32 sealed pairs	17 sealed pairs	denominators of 10 of 17 arms (4.5 of 8 trials)

and the disagreement is recorded as an erratum. On anchor 1 that verification had produced a quotation-backed record of 11 source-confirmed discrepancies in the published review (swapped arms, a mislabeled outcome row, analysed counts presented as randomized). On anchor 2 the published cells reproduced cleanly. On anchor 3 the review prints no per-trial cells in text: layer 1 was read from its raster forest plot and verified by reconstruction (the 16 read cells predict the 48 other values printed on the figure, all of which reproduce), and layer 2 found two trials wrong at the source. For Shaker 2025 the review records 15/30 in the methylene-blue arm; the trial has three arms (placebo 14/30; methylene blue 9/30 and 6/30), so the published numerator sums both active arms while the denominator keeps one, and layer 2 holds 15/60 against 14/30 under the Handbook’s combined-arms rule [Higgins et al., 2023]. For Aguilar 2016 the review records 24/30 and 19/30; the trial reports mortality of 20.0% and 36.6% in arms of 30, i.e. 6 and 11 deaths, so the published cells are the survivors, and layer 2 holds 6/30 against 11/30. Three internal inconsistencies of the review were also frozen before the runs: its primary analysis is described three incompatible ways (the forest plot governs, as the only description with cells), its unnamed random-effects estimator reproduces the printed diamond only as Paule–Mandel with the Hartung–Knapp adjustment [Veroniki et al., 2016, IntHout et al., 2014], and Aguilar contributes a 21-day figure to an analysis labeled 28–30 days (used as the review used it, with its exclusion run as a registered sensitivity). Grading was against layer 2 throughout, so no verdict rests on the published 0.73. The same ruler applied to layer 1 yields a *human row* for each anchor: the published review’s own cells scored against the sources.

2.3 The perturbation reading proof

Before any model read a trial, a sealed deterministic operator displaced load-bearing numbers in its text (anchors 1 and 2: 32 and 17 sealed value pairs). A model that returns a displaced value read the text in front of it; one that returns the original published value is reciting. The models operated only in the perturbed world; the original↔perturbed maps were held by the graders and applied after extraction as a pure table lookup (the reversal lens). Anchor 3’s cells are integer counts, which cannot be displaced without colliding with other small integers in

the text, so its proof was placed on the per-arm denominators of 10 of 17 arms, with every dependent percentage recomputed for internal coherence and the trial totals moved with their arms, in figures and in words; the event counts carry no proof, three trials none at all, and the coverage — four and a half of eight trials — is stated beside every anchor-3 result. Anchor 3 used a boundary-aware lens of its own, because the frozen substring lens commits 42 undue substitutions on its two-digit seal values. A mechanical screen flagged as a recitation candidate any raw cell containing an original sealed value while the displaced image the model read was absent; every candidate was adjudicated against the perturbed text. One costless trap was registered in advance: a sheet returning the review’s 24/30 for Aguilar has reproduced the review, not read the trial.

2.4 Models, hardware and instruments

Six open-weight models ran as peers: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`, all Q4_K_M builds (about 4.5 bits per weight), and a sixth, `smtek/Qwen3.8-27B:Q2_K_XL`, a 27.3-billion-parameter mixed-precision build whose bit depth was measured from its weight file as 3.13 bits per weight, because inference quality below four bits depends on the scheme [Dettmers and Zettlemoyer, 2023]. All calls ran on one consumer mini-computer (AMD Ryzen 7, Radeon 780M integrated GPU, 32 GB RAM, no discrete GPU) served by Ollama with a 24,576-token context, reasoning channels off, vendor default sampling, no seed, one model resident at a time; no cloud model was called in any evaluated pipeline. Two frozen English extraction sheets ran on every anchor: v1 asks for each cell a value and a locator naming the section or table it came from; v2 additionally requires a verbatim source quotation for every cell (output allowance 4,000 and 8,000 tokens). Anchor 1’s sheet covers descriptive and outcome fields per trial; anchor 2’s records the change in HbA1c, its dispersion and the arm sizes; anchor 3’s asks, for every arm the trial has, the event count, the denominator, the reported percentage and the outcome timepoint, with no polarity instruction and no multi-arm reduction, which the frozen corrector performs.

2.5 Extraction, grading, pooling and adjudication

Phase 1 ran $6 \times (14 + 7 + 8) \times 2 \times 2 = 696$ calls: two replicates per sheet, over the perturbed corpora, through a resumable harness that accepted an output only if it parsed with the expected keys and had not stopped on the token ceiling, recording token counts, stop reason and duration per call. Phase 2 reversed the seal on each sheet and compared every cell with layer 2: the frozen magnitude-based comparator on anchors 1 and 2 (124 and 49 eligible cells per sheet, those with a digit-bearing source-verified value), exact integer comparison on anchor 3 (32 cells: the event count and denominator of 16 arms); replicate stability applied the same comparator between the two replicates. Phase 3 pooled each sheet’s own cells with frozen engines and no model in the loop: DerSimonian–Laird [DerSimonian and Laird, 1986] risk ratios on anchor 1 over the review’s per-outcome rosters, mean differences on anchor 2 after a deterministic route selector (declared change used directly; standard errors and confidence intervals converted to standard deviations; baseline-and-final pairs combined under the $r = 0.5$ rule [Higgins et al., 2023]), and odds ratios on anchor 3 after the frozen multi-arm reduction; on anchor 3 the review’s own estimator (Paule–Mandel with Hartung–Knapp) was run by the comparator only, to measure the review. Tolerances were ± 0.01 on ratios and interval limits and ± 0.1 on mean differences. Provenance nets ran over every v2 cell as a declared exploratory arm: one flags a quotation not found in the source, the other a value not found in its quotation.

Phase 4 adjudicated every non-matching cell with the source quotation before the verdict. Named mechanical rules classified each cell by mechanism (omission; faithful in another encoding or population layer; reading slip; grader-side artifact); the residue was decided by the reader — the assistant, under the author’s delegation, every verdict reversible — and all 85

reader verdicts plus the 39 cells that put the key itself on trial were re-read by three independent adversarial reviewers with the sources open (ten disputed, nine corrections accepted after re-verification). The adjudicated score adds to the mechanical score only cells in which the model was right and the grader wrong; faithful re-encodings, population-layer choices and summaries stay divergent. The complete cell-by-cell record, with the quotation and the rule or reader that decided each cell, is in the repository.

2.6 Software and analysis

All computation used Python 3.12 and the repository’s validated engine functions (risk ratio, mean difference and odds ratio with intervals; standard-deviation conversions; DerSimonian–Laird and Paule–Mandel/Hartung–Knapp pooling); no external statistics package was used. Replicates measure stability, not significance; with one corpus per anchor, comparisons between models are descriptive, and the pre-registered hypotheses were read against their fixed criteria.

3 Results

3.1 The campaign and the pre-registered hypotheses

All 696 calls completed and passed the integrity gate, with no refusal and no truncation, in 39.7 h of summed call time (3.42 min per call; 7.52 million input and 0.78 million output tokens; largest prompt 17,898 of 24,576 tokens). The mechanical recitation screen raised 13 candidate cells, all on anchor 1, and all 13 resolved to reconstruction from values the operator had left in the perturbed text (a surviving percentage beside a displaced count, a surviving prose restatement, surviving counts under a displaced percentage); no attributable recitation occurred, and the anchor-3 trap (24/30) appeared on no sheet. Table 2 lists the six hypotheses and their outcomes.

Table 2: Pre-registered hypotheses and outcomes.

	Criterion, fixed before the runs	Result	Verdict
H12.1	every anchor-3 sheet parses and its pool is digit-consistent with its own cells (± 0.01)	pools consistent on 12 of 12 pairs; three v2 sheets refused by the strict JSON reader	9 of 12 (12 of 12, tolerant reader)
H12.2	≥ 1 of 6 models writes Aguilar 6/30 (not 24/30) and reduces Shaker to 15/60	v1: Aguilar 6/30 in 4 of 6, Shaker 15/60 in 6 of 6, both in 4; errata reproduced by 0	met
H12.3	v2 costs ≥ -4 cells of 124 on anchor 1 <i>and</i> pools closer to layer 2 in ≥ 4 of 6 models on ≥ 2 anchors	mean -6.8 cells (range -1 to -20); closer on 0 of 3 anchors	both fail
H12.4	(exploratory) provenance nets over-fire on the table-dwelling anchor 3	19–39% and 7–43% of anchor-3 cells vs 10–57% and 7–45% elsewhere	no excess
H12.5	context control: gemma4:12b anchor-1 v1 within 103 ± 4 cells and anchor-2 pool within ± 0.05 of -0.27	109/124 (108 mechanical); pool -0.27	one cell outside
H12.6	bounded null: no model above 50% of the 32 anchor-3 cells	11 of 12 pairs above 50%	null falls

3.2 Cells

Cell agreement with the source-verified key (Table 3) ranged, per model and sheet, from 85/124 to 111/124 on anchor 1, from 16/42 to 35/49 on anchor 2 and from 16/28 to 32/32 on anchor 3; the 27B build held the top score on anchors 1 and 2 under both sheets, and two 14B models, **qwen3:14b** and **deepseek-r1:14b**, scored 32 of 32 on anchor 3 under v1. On the same ruler the published reviews’ own cells scored 102/124, 49/49 and 29/32: on anchor 1, 12 of the review’s

22 non-matching cells were its source-confirmed discrepancies; on anchor 3 they were the three cells of the two errata. Replicate agreement ranged from 75.9% to 98.3% (anchor 1), 59.5% to 100% (anchor 2) and 62.5% to 100% (anchor 3). Under v2 every model scored the same or lower on anchor 1 (mean -6.8 cells, range -1 to -20), and three v2 sheets on anchor 3 were refused by the strict JSON reader because the mandatory quotation carried control characters inherited from PDF-derived text (a tolerant reader, run as a declared sensitivity, recovered them to 28/32, 20/32 and 30/32). The provenance nets flagged 19–39% of eligible anchor-3 cells on the quotation-exists net and 7–43% on the value-in-quotation net, within the ranges seen on anchors 1 and 2 (10–57% and 7–45%).

Table 3: Cells agreeing with the source-verified key, by model, sheet and anchor (adjudicated scores). First row: the published review’s own cells on the same ruler. †v2 sheets refused by the strict JSON reader; denominator reduced.

Extraction	Anchor 1 (124 cells)		Anchor 2 (49)		Anchor 3 (32)	
	v1	v2	v1	v2	v1	v2
Published (human)		102		49		29
gemma4:12b	109	106	33	31	23	23
qwen3:14b	105	85	24	21	32	24/28†
llama3.1:8b	92	86	24	16/42	26	23
qwen3.5:9b	95	91	30	25	28	16/28†
deepseek-r1:14b	109	102	25	24	32	22/24†
qwen3.8:27b	111	110	35	31	28	28

3.3 Pooled estimates on anchors 1 and 2

From each sheet’s own cells the frozen engines built anchor 1’s pooled morbidity risk ratio between 0.738 and 0.788 against the source-verified 0.787, nine of twelve pairs within 0.01 and eleven within 0.02, every interval crossing 1 as published (Table 4); pooled mortality lay between 0.980 and 1.078 on eleven pairs, and one pair (the 27B build, v1) fell to 0.400 [0.018; 9.121] through a single stray denominator, a figure-caption number written as one trial’s control-arm size. On anchor 2, five of twelve pairs landed within the pre-registered ± 0.1 of the source-verified -0.24 : **gemma4:12b** v1 at -0.27 [-0.38 ; -0.17] and the 27B build v1 at -0.25 over all seven trials, and three pools over three or four trials at -0.20 to -0.22 ; the other seven pairs landed between -0.42 and -0.88 , with two to seven trials pooled (Table 4). Eleven anchor-2 cells were adjudicated in the models’ favor because the lens could not reverse what the model had correctly read (Section 3.5); with those cells at the value the lens should have returned, six pools move toward the source (**qwen3:14b** v1 from -0.42 to -0.32 , **deepseek-r1:14b** v1 from -0.46 to -0.27 , **qwen3.5:9b** v2 from -0.88 to -0.32 , the other three by 0.14 to 0.25), a sensitivity recorded beside the primary that changes no hypothesis verdict. Under v2 the pool was closer to layer 2 than under v1 in one model on anchor 1, two on anchor 2 and none on anchor 3, against H12.3’s criterion of four or more on two anchors.

3.4 Anchor 3: the corrected meta-analysis

No sheet of any model reproduced either erratum: zero cells agreed with the published 24/30 or 15/30 against the source (Table 5). Under v1, four of six models read Aguilar as 6/30 from the Spanish primary, and the other two returned no mortality entry for that trial on either sheet (graded as omissions); six of six enumerated Shaker’s three arms, so that the frozen reduction yielded 15/60; four of six landed both cells. **qwen3:14b** and **deepseek-r1:14b** scored 32 of 32 cells and pooled, under the campaign engine, exactly to the source-verified constant, OR 0.484

Table 4: Pooled estimates built by the frozen engines from each sheet’s own cells, anchors 1 and 2 (estimate [95% CI]; anchor 2 with the number of trials pooled). ‡one stray denominator.

Extraction	Sheet	A1 morbidity RR	A1 mortality RR	A2 HbA1c MD (trials)
Published (human)	—	0.778 [0.567; 1.068]	1.021 [0.446; 2.337]	−0.24 [−0.32; −0.16] (7)
Layer 2 (source)	—	0.787 [0.580; 1.068]	1.023 [0.446; 2.346]	−0.24 [−0.32; −0.16] (7)
gemma4:12b	v1	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	−0.27 [−0.38; −0.17] (7)
	v2	0.779 [0.569; 1.065]	1.078 [0.470; 2.471]	−0.21 [−0.23; −0.19] (4)
qwen3:14b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	−0.42 [−0.91; 0.06] (7)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	−0.22 [−0.59; 0.15] (3)
llama3.1:8b	v1	0.738 [0.510; 1.070]	1.023 [0.447; 2.344]	−0.63 [−1.03; −0.24] (5)
	v2	0.767 [0.552; 1.066]	1.023 [0.447; 2.344]	−0.58 [−0.85; −0.31] (5)
qwen3.5:9b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	−0.63 [−1.07; −0.19] (5)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	−0.88 [−2.14; 0.39] (3)
deepseek-r1:14b	v1	0.788 [0.579; 1.072]	0.980 [0.428; 2.246]	−0.46 [−0.66; −0.25] (5)
	v2	0.769 [0.558; 1.061]	1.021 [0.446; 2.337]	−0.52 [−0.83; −0.21] (2)
qwen3.8:27b	v1	0.787 [0.580; 1.068]	0.400 [0.018; 9.121]‡	−0.25 [−0.37; −0.12] (7)
	v2	0.779 [0.569; 1.065]	1.023 [0.446; 2.346]	−0.20 [−0.34; −0.05] (4)

[0.319; 0.735], and under the review’s own estimator to 0.484 [0.342; 0.685], where the review published 0.73 [0.40; 1.36]; the review’s own cells scored 29 of 32, and only those two sheets scored above them. Every pair’s DerSimonian–Laird pool lay between 0.418 and 0.548 with the upper limit below 1; under the review’s estimator one upper limit reached 1.002 (**gemma4:12b** v1, six trials). Of the 63 non-matching anchor-3 cells, 56 were omissions — five sheets wrote nothing for the Spanish trial and four for the trial that publishes no per-arm count — and the seven reading slips were two survivor counts reported as deaths, two percentages written in the denominator slot and three cause-specific partial counts. By the review’s own estimator, its published diamond becomes 0.60 [0.33; 1.12] with the Shaker cell corrected, 0.59 [0.39; 0.91] with the Aguilar cell corrected, 0.48 [0.34; 0.69] with both, and 0.49 [0.33; 0.74] with both corrected and Aguilar removed.

Table 5: Anchor 3: the two cells that decide (methylene-blue arm of Aguilar 2016 and of Shaker 2025), the 32-cell score and the pooled odds ratio under the campaign engine (DerSimonian–Laird, trials pooled in parentheses) and under the review’s own estimator (Paule–Mandel with Hartung–Knapp). NR: no mortality entry returned. ‡as in Table 3.

Extraction	Sheet	Aguilar MB	Shaker MB	Cells	OR, DL (trials)	OR, PM–HK
Published (human)	—	24/30	15/30	29/32	0.737 [0.449; 1.208] (8)	0.73 [0.40; 1.36]
Layer 2 (source)	—	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
gemma4:12b	v1	NR	15/60	23/32	0.519 [0.321; 0.839] (6)	0.520 [0.270; 1.002]
	v2	NR	15/60	23/32	0.418 [0.246; 0.711] (6)	0.418 [0.211; 0.828]
qwen3:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	24/28‡	0.481 [0.311; 0.745] (6)	0.481 [0.393; 0.589]
llama3.1:8b	v1	6/30	15/60	26/32	0.496 [0.318; 0.773] (7)	0.496 [0.319; 0.770]
	v2	6/30	NR	23/32	0.517 [0.322; 0.832] (6)	0.517 [0.348; 0.770]
qwen3.5:9b	v1	6/30	15/60	28/32	0.460 [0.299; 0.708] (7)	0.460 [0.325; 0.651]
	v2	NR	15/60	16/28‡	0.450 [0.267; 0.758] (4)	0.450 [0.212; 0.951]
deepseek-r1:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	22/24‡	0.548 [0.362; 0.831] (6)	0.548 [0.388; 0.775]
qwen3.8:27b	v1	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]
	v2	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]

3.5 The 633 non-matching cells

Adjudication classified the 633 cells that failed the layer-2 comparison (297 on anchor 1, 273 on anchor 2, 63 on anchor 3; Table 6). Three hundred and thirty-one were faithful to the source in another encoding or population layer: 165 population-layer choices (an analysed count where the key holds the randomized one, or the reverse), 78 case-mix summaries, 54 percent-for-count re-encodings, 16 standard deviations derived by the key’s own rule from the analysed n the model read, 8 unit re-encodings (mmol/mol), 6 cells for which the primary prints two contradictory counts, 3 partial re-encodings and 1 “not reported” written in words. One hundred and seventy-five were omissions. One hundred and six were reading slips, 74 of them on the continuous anchor: 52 values taken from another row or scope (a between-group difference written as an arm’s change, another visit, another outcome’s analysed count, a factorial cell for a margin), 28 dispersions of the wrong type, 6 arm swaps, 2 polarity inversions, 2 field mixes and 16 values the text does not print — eight ratio-style strings and one count reconstructed from a percentage and an inconsistently printed n , all from `llama3.1:8b` on anchor 1, and seven computed dispersions, change means and one denominator on anchor 2 from five models; none reached anchor 3. Twenty-one cells were grader-side artifacts and were credited to the models: ten copies of one seal-pair substring collision inside a correctly transcribed value, one displaced value written with a Unicode minus that the lens patterns miss, and ten cells in which a model applied the key’s own derivation rule (change = final – baseline) to the displaced text and landed on a value the lens had never seen. Thirty-nine cells put the key itself on trial — three trials that print both the randomized and the analysed count under a field named *n randomized*, where the key holds the layer the review pooled — and were listed for the author without flipping.

Table 6: Verdicts on the 633 non-matching cells, by anchor.

Verdict	Anchor 1	Anchor 2	Anchor 3	All
Faithful to the source, other encoding or layer	197	134	0	331
Omission	65	54	56	175
Reading slip	25	74	7	106
Model correct, grader wrong	10	11	0	21
Non-matching cells	297	273	63	633

4 Discussion

Six small local models, reading perturbed copies of 29 trials on a consumer machine, rebuilt two published meta-analyses to their published estimates and the third as its sources support it rather than as published. Under the perturbation control none of that is attributable to memory of the reviews, and under the source-verified key the published reviews were graded on the same ruler as the models.

The third anchor is the decisive result. Its published diamond can be reproduced only by reproducing two transcription errors, and no sheet of any model did: four models read the Spanish primary as the source prints it, all six enumerated the three-arm trial, and two pooled to the corrected estimate to the third decimal — the mortality benefit the review’s own Table 1 records and its forest plot denies. Three things scope that result, and the protocol wrote them down before the first call: the errata were raised by this study’s adjudicator during key construction, so the models were measured against them rather than credited with finding them; the anchor-3 sheet asks four fields per arm over enumerated arms, the two axes on which the human extractors failed, so the instrument helps a reader land the correction; and

the event counts carry no reading proof — it rests on displaced denominators in four and a half of eight trials — while the primaries, unlike the two post-cutoff reviews, fall inside the models’ training windows, so a recited death count cannot be excluded, although the recited review cell that would have betrayed recall of the meta-analysis never appeared. What the anchor establishes is narrower and useful: a source-faithful local reader produces the corrected meta-analysis without any instruction to look for errors, and a benchmark grading against the published cells would have scored its two best readers zero on the two cells that matter.

The division of difficulty between the anchors appears structural. Dichotomous material is transcription — counts and denominators printed redundantly across table, prose and percentage — and the pipeline dampens what error remains, because a risk ratio shifts numerator and denominator together when a model takes another population layer: twelve sheets took visibly different denominator routes and still landed within 0.05 of the source-verified morbidity pool. Each continuous quantity is printed once, in one of several competing layers, and the decisive step is semantic — identifying what a printed dispersion *is*; a mislabeled type sends the conversion down the wrong route, and inverse-variance pooling amplifies the error because weight scales with $1/SD^2$ [Higgins et al., 2023]. The same architecture is therefore error-dampening for count outcomes and error-amplifying for continuous ones, which gives the uneven accuracy of the feasibility literature [Schmidt et al., 2024, Khraisha et al., 2024] a structure: here the unevenness tracks the outcome type at least as much as the model. The ranking of models was anchor-dependent in the same way: the 27B build read best on the two layered anchors and two 14B builds on the count anchor — an observation, not a tested hypothesis.

None of the reading failures was a fabrication: every divergent cell traces to the text read or to a computation over it, and the inventory — omissions, population-layer choices, format variants, mislabeled quantities — is systematic and auditable. This is consistent with a cloud-model extraction study in which errors were few and *missed data was the most common error type* [Gartlehner et al., 2024], so omission dominating fabrication at opposite ends of the size and deployment spectrum suggests a property of the task rather than of one model class. The dispersion-type slips that concentrate on the continuous anchor are the class an audit of a published meta-analysis found in 9 of 16 human-extracted effect sizes, where correcting the standard-error/standard-deviation mix-up moved the pooled estimate from 1.46 to 0.54 [Sandercock, 2024]; it deserves a dedicated sheet-versus-source dispersion check rather than a better model. The population-layer choices, the largest faithful class, are likewise a human ambiguity: on one trial every sheet took the completers layer the paper headlines while the review had pooled the intention-to-treat layer, and both are printed.

The human rows place the published reviews on the models’ ruler: 102 of 124, 49 of 49 and 29 of 32 cells, so the first review sits below the cast’s better readers and above its weakest, and the third below its two best. That the human column carries errors at all is consistent with the methodological literature rather than remarkable within it [Gøtzsche et al., 2007, Jones et al., 2005, Tendal et al., 2011, Mathes et al., 2017]; what this study adds is the quotation trail cell by cell and a model column audited beside the human one. The errors with conclusion-level weight on either side — survivors entered as deaths, a summed numerator over one arm’s denominator, a dispersion of the wrong type — are mislabels, suggesting that label semantics, not numbers, is where extraction fails for both kinds of extractor and where verification effort should concentrate. The same quotation rule protected the models from the graders — 21 cells charged by the instruments were credited back, ten of them for applying the key’s own derivation rule to the displaced text — and is the structural constraint on the plausibility bias any judge, human or model, exhibits [Zheng et al., 2023].

The mandatory-quotation sheet answered its question in the negative at the scale the question asked for: it cost accuracy where the source prints table rows rather than sentences, most for the model that follows the instruction most literally, brought no pooled estimate closer to the source, and carried the source’s control characters into three sheets the strict reader refused.

Its value — a reviewer can adjudicate a cell without opening the article — is a deployment decision, not a benchmark result, and v2 is not adopted as the default. The provenance nets fired no more on the table-dwelling anchor than on the others, so the table-cell blind spot belongs to the quotation, not to the nets. The context control fell one cell outside its band, upward, on the descriptive anchor only; no comparison with the earlier development record is made here.

Three design choices on anchor 3 are the author’s to disagree with, and each is stated so that disagreeing requires no reconstruction. The reading proof displaces denominators with coherent recomputation rather than percentages alone, which would make count, denominator and percentage contradict one another inside the source, or than leaving the anchor unperturbed; it buys a partial proof whose limits are printed beside every anchor-3 number. The forest plot governs among the review’s three incompatible descriptions of its primary analysis because it is the only one with cells attached. Aguilar’s 21-day figure is pooled into an analysis the review labels 28–30 days because that is what the review did, with its exclusion run as a registered sensitivity: 0.484 with the trial, 0.493 without, the conclusion unchanged either way.

Several limitations bound these claims. Three anchors and 29 primaries are the whole evidence base, two of the anchors from one journal family; every superlative is scoped to these corpora, and no between-model inference is attempted. One quantization per model and one hardware profile were tested, so parameter count and precision are not separable. On anchor 3 the reading proof covers the denominators of four and a half trials and no event count, the primaries predate the models’ training cutoffs, the errata were raised by the adjudicator before any model read the anchor, and the sheet is built on the axes where the human extractors failed. The adjudication of the 633 cells was delegated by the author to the assistant under the quotation rite, which trades independence for completeness; the independent re-read that reversed nine of the reader’s 85 verdicts mitigates but does not remove that. The human rows score the published cells against a key built by verifying those same cells, so their divergences are the verification record by construction. The mechanical comparator approximates the adjudicated ruler, so absolute cell scores understate agreement. Finally, the perturbation operator’s residual gaps — surviving percentages, totals and prose restatements — are documented and exploitable, though what they produced here was reconstruction, not recall.

In conclusion, one consumer machine running a well-chosen local reader — a 12-to-14-billion-parameter model at 4.5 bits per weight, or 27 billion at 3.13 — with frozen deterministic engines rebuilt three published meta-analyses from their primary trials as the sources support them: two at their published estimates, and one at a corrected estimate whose conclusion differs from the published one. The next question is how often grading against the primaries, rather than against the review, would change published estimates if applied at scale.

Data and code availability

The pre-registered protocol with its review, the frozen English instruments, the perturbation seals with SHA-256 fingerprints, the two-layer answer keys with their source-verification records, the anchor-3 errata record and held alert table, the deterministic engines and graders, the campaign harness, all 696 raw model outputs with per-call records, the Phase-2 and Phase-4 records with the 633 adjudicated cells and their quotations, the review of the reader’s verdicts, and the exact quantized builds pinned by Ollama digest are public at <https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr> and archived at Zenodo [Barra, 2026]. Paywalled primaries are not redistributed; scripts rebuild that corpus stratum from legally obtained copies.

Declarations

Funding. No external funding. All computation ran on the author’s own consumer hardware.

Competing interests. The author declares no competing interests. The AI assistant declared below was used under an ordinary consumer subscription paid by the author, who has no financial or professional relationship with Anthropic or with the organizations whose open-weight models are evaluated here.

Use of artificial intelligence. The study was designed and conducted by the author with the assistance of Claude (Anthropic), which drafted the protocol from the author’s design decisions (three assistant-originated choices were put to the author before any run), built the anchor-3 key, perturbation instrument and harness, implemented the graders and adjudication rules, performed the Phase-4 adjudication under the author’s delegation with every verdict quotation-bound and reversible, and drafted this manuscript under the author’s direction and review. The evaluated local models produced only the extraction sheets graded here; every downstream number was produced by frozen code and verified mechanically. Per the position of the evidence-synthesis organizations [Flemyng et al., 2025], every AI contribution is reported and the author retains responsibility for the work.

Ethics approval and consent. Not applicable: no human participants and no new data collection; only published aggregate reports were analyzed.

Acknowledgments

I thank my family, for the support that made this work possible; my teachers and professors, whose lessons are present throughout it; and Sabin Diagnóstico e Saúde, where I work, for the daily practice of diagnostic medicine in which the habits of measurement behind this project were formed.

References

- M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, and D. Moher. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021. doi: 10.1136/bmj.n71.
- R. Borah, A. W. Brown, P. L. Capers, and K. A. Kaiser. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open*, 7(2):e012545, 2017. doi: 10.1136/bmjopen-2016-012545.
- L. Schmidt, K. Hair, S. Graziosi, F. Campbell, C. Kapp, A. Khanteymoori, D. Craig, M. Engelbert, and J. Thomas. Exploring the use of a large language model for data extraction in systematic reviews: a rapid feasibility study. In *Proceedings of the 3rd Workshop on Augmented Intelligence for Technology-Assisted Reviews Systems (ALTARS)*, 2024. arXiv:2405.14445.
- Q. Khraisha, S. Put, J. Kappenberg, A. Warraitch, and K. Hadfield. Can large language models replace humans in systematic reviews? Evaluating GPT-4’s efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages. *Research Synthesis Methods*, 15(4):616–626, 2024. doi: 10.1002/jrsm.1715.

- A. Konet, I. Thomas, G. Gartlehner, L. Kahwati, R. Hilscher, S. Kugley, K. Crotty, M. Viswanathan, and R. Chew. Performance of two large language models for data extraction in evidence synthesis. *Research Synthesis Methods*, 15(5):818–824, 2024. doi: 10.1002/jrsm.1732.
- D. C. W. Lee, K. M. O’Brien, J. Presseau, S. Yoong, C. Lecathelinais, L. Wolfenden, J. Thomas, A. Arno, B. Hutton, and R. K. Hodder. Accuracy and efficiency of using artificial intelligence for data extraction in systematic reviews. a noninferiority study within reviews. *medRxiv preprint*, 2026. doi: 10.64898/2026.02.25.26347053.
- C. Xie, W. Kong, L. Pi, D. Qi, Y. Yang, B. Wang, and H. Liang. Performance of large language models in automated medical literature screening: A systematic review and meta-analysis. *Journal of Evidence-Based Medicine*, page e70166, 2026. doi: 10.1111/jebm.70166.
- L. Bakker, T. Caganek, A. Rooprai, and S. Hume. Systematic reviews in minutes to hours using artificial intelligence. *medRxiv preprint*, 2026. doi: 10.64898/2026.02.06.26345764.
- L. McLaughlin, M. S. Walz, and C. Arries. Fully automated systematic review generation via large language models: Quality assessment and implications for scientific publishing. *medRxiv preprint*, 2026. doi: 10.64898/2026.02.18.26346559.
- E. Flemyng, A. Noel-Storr, B. Macura, G. Gartlehner, J. Thomas, J. J. Meerpohl, Z. Jordan, J. Minx, A. Eisele-Metzger, C. Hamel, P. Jemioło, K. Porritt, and M. Grainger. Position statement on artificial intelligence (AI) use in evidence synthesis across Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence 2025. *Environmental Evidence*, 2025. doi: 10.1186/s13750-025-00374-5.
- J. Gallifant, M. Afshar, S. Ameen, Y. Aphinyanaphongs, S. Chen, G. Cacciamani, D. Demner-Fushman, D. Dligach, R. Daneshjou, C. Fernandes, L. H. Hansen, A. Landman, L. Lehmann, L. G. McCoy, T. Miller, A. Moreno, N. Munch, D. Restrepo, G. Savova, R. Umeton, J. W. Gichoya, G. S. Collins, K. G. M. Moons, L. A. Celi, and D. S. Bitterman. The TRIPOD-LLM reporting guideline for studies using large language models. *Nature Medicine*, 31(1):60–69, 2025. doi: 10.1038/s41591-024-03425-5.
- I. C. Wiest, D. Ferber, J. Zhu, M. van Treeck, S. K. Meyer, R. Juglan, Z. I. Carrero, D. Paech, J. Kleesiek, M. P. Ebert, D. Truhn, and J. N. Kather. Privacy-preserving large language models for structured medical information retrieval. *npj Digital Medicine*, 7:257, 2024. doi: 10.1038/s41746-024-01233-2.
- D. J. Spaanderman, K. Prathaban, P. Zelina, K. Mouheb, L. Hejtmánek, M. Marzetti, A. W. Schurink, D. Chan, R. Niemantsverdriet, F. Hartmann, Z. Qian, M. G. J. Thomeer, P. Holub, F. Akram, F. J. Wolters, M. W. Vernooij, C. Verhoef, E. E. Bron, V. Nováček, D. J. Grünhagen, W. J. Niessen, M. P. A. Starmans, and S. Klein. Evaluating open-weight large language models for structured data extraction from narrative medical reports across multiple use cases and languages. *arXiv preprint*, 2025. arXiv:2511.10658.
- N. Carlini, D. Ippolito, M. Jagielski, K. Lee, F. Tramèr, and C. Zhang. Quantifying memorization across neural language models. arXiv:2202.07646v3, 2023. arXiv preprint.
- C. Xu, S. Guan, D. Greene, and M.-T. Kechadi. Benchmark data contamination of large language models: A survey. *arXiv preprint*, 2024. arXiv:2406.04244.
- S. Golchin and M. Surdeanu. Time travel in LLMs: tracing data contamination in large language models. arXiv:2308.08493v3, 2024. International Conference on Learning Representations (ICLR) 2024.

- P. C. Gøtzsche, A. Hróbjartsson, K. Maric, and B. Tendam. Data extraction errors in meta-analyses that use standardized mean differences. *JAMA*, 298(4):430–437, 2007. doi: 10.1001/jama.298.4.430.
- A. P. Jones, T. Remington, P. R. Williamson, D. Ashby, and R. L. Smyth. High prevalence but low impact of data extraction and reporting errors were found in Cochrane systematic reviews. *Journal of Clinical Epidemiology*, 58(7):741–742, 2005. doi: 10.1016/j.jclinepi.2004.11.024.
- B. Tendam, J. P. T. Higgins, P. Jüni, A. Hróbjartsson, S. Trelle, E. Nüesch, S. Wandel, A. W. Jørgensen, K. Gesser, S. Ilsøe-Kristensen, and P. C. Gøtzsche. Disagreements in meta-analyses using outcomes measured on continuous or rating scales: observer agreement study. *BMJ*, 339:b3128, 2009. doi: 10.1136/bmj.b3128.
- B. Tendam, E. Nüesch, J. P. T. Higgins, P. Jüni, and P. C. Gøtzsche. Multiplicity of data in trial reports and the reliability of meta-analyses: empirical study. *BMJ*, 343:d4829, 2011. doi: 10.1136/bmj.d4829.
- T. Mathes, P. Klafen, and D. Pieper. Frequency of data extraction errors and methods to increase data extraction quality: a methodological review. *BMC Medical Research Methodology*, 17(1):152, 2017. doi: 10.1186/s12874-017-0431-4.
- G. B. Barra. EXTRAI: an evidence-extraction benchmark for local LLMs (Brazilian Portuguese instructions). GitHub repository and Zenodo archive, version 1.2.0, 2026. DOI 10.5281/zenodo.22159050 (concept); 10.5281/zenodo.22190474 (v1.2.0).
- N. Ashraf, O. U. U. Zargar, and A. Albina. Comparison of goal-directed fluid therapy and conventional fluid therapy in elective major abdominal surgery: A meta-analysis of randomized controlled trials. *Cureus*, 2026. doi: 10.7759/cureus.110243.
- Y. Khraise. Effect of low-carbohydrate diets on glycemic control in type 2 diabetes mellitus: A systematic review and meta-analysis of randomized controlled trials. *Cureus*, 2026. doi: 10.7759/cureus.108479.
- D. R. Rodríguez-Lima, A. Rodríguez-Villegas, J. D. Toro Egas, and E. M. C. Bautista. The effectiveness of methylene blue in adult shock: A systematic review, meta-analysis, and trial sequential analysis of randomized controlled trials. *Journal of Clinical Medicine*, 15(12):4481, 2026. doi: 10.3390/jcm15124481.
- J. P. T. Higgins, J. Thomas, J. Chandler, M. Cumpston, T. Li, M. J. Page, and V. A. Welch, editors. *Cochrane Handbook for Systematic Reviews of Interventions, version 6.4 (updated 2023)*. Cochrane, 2023. Available from <https://training.cochrane.org/handbook>.
- A. A. Veroniki, D. Jackson, W. Viechtbauer, R. Bender, J. Bowden, G. Knapp, O. Kuss, J. P. T. Higgins, D. Langan, and G. Salanti. Methods to estimate the between-study variance and its uncertainty in meta-analysis. *Research Synthesis Methods*, 7(1):55–79, 2016. doi: 10.1002/jrsm.1164.
- J. IntHout, J. P. A. Ioannidis, and G. F. Borm. The Hartung-Knapp-Sidik-Jonkman method for random effects meta-analysis is straightforward and considerably outperforms the standard DerSimonian-Laird method. *BMC Medical Research Methodology*, 14:25, 2014. doi: 10.1186/1471-2288-14-25.
- T. Dettmers and L. Zettlemoyer. The case for 4-bit precision: k -bit inference scaling laws. arXiv:2212.09720v2, 2023. arXiv preprint.

- R. DerSimonian and N. Laird. Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3): 177–188, 1986. doi: 10.1016/0197-2456(86)90046-2.
- G. Gartlehner, L. Kahwati, R. Hilscher, I. Thomas, S. Kugley, K. Crotty, M. Viswanathan, B. Nussbaumer-Streit, G. Booth, N. Erskine, A. Konet, and R. Chew. Data extraction for evidence synthesis using a large language model: a proof-of-concept study. *Research Synthesis Methods*, 15(4):576–589, 2024. doi: 10.1002/jrsm.1710.
- G. Sandercock. The standard error/standard deviation mix-up: potential impacts on meta-analyses in sports medicine. *Sports Medicine*, 54(6):1723–1732, 2024. doi: 10.1007/s40279-023-01989-9.
- L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. arXiv:2306.05685.